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User equipment behavior and requirements for positioning measurement without gap

Abstract

Various embodiments herein provide techniques for positioning measurements in a wireless cellular network without a measurement gap. The measurements may be performed in one or more positioning frequency layers (PFLs). Additionally, techniques for activation and/or deactivation of a measurement gap are provided. Other embodiments may be described and claimed.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application claims priority to U.S. Provisional Patent Application No. 63/297,646, which was filed Jan. 7, 2022; the disclosure of which is hereby incorporated by reference.

FIELD

(1) Various embodiments generally may relate to the field of wireless communications. For example, some embodiments may relate to user equipment (UE) behavior and requirements for positioning measurements without a gap.

BACKGROUND

(2) A user equipment (UE) in a wireless cellular network typically receives positioning reference signals (PRSs) from one or more next generation Node Bs (gNBs). The UE performs positioning measurements on the PRSs, such as a received signal time difference (RSTD) measurement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments will be readily understood by the following detailed description in conjunction with the accompanying drawings. To facilitate this description, like reference numerals designate like structural elements. Embodiments are illustrated by way of example and not by way of limitation in the figures of the accompanying drawings.

(2) FIG. 1 schematically illustrates a wireless network in accordance with various embodiments.

(3) FIG. 2 schematically illustrates components of a wireless network in accordance with various embodiments.

(4) FIG. 3 is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein.

(5) FIG. 4 illustrates an example process for practicing the various embodiments herein.

(6) FIG. 5 illustrates another example process for practicing the various embodiments herein.

DETAILED DESCRIPTION

(7) The following detailed description refers to the accompanying drawings. The same reference numbers may be used in different drawings to identify the same or similar elements. In the following description, for purposes of explanation and not limitation, specific details are set forth such as particular structures, architectures, interfaces, techniques, etc. in order to provide a thorough understanding of the various aspects of various embodiments. However, it will be apparent to those skilled in the art having the benefit of the present disclosure that the various aspects of the various embodiments may be practiced in other examples that depart from these specific details. In certain instances, descriptions of well-known devices, circuits, and methods are omitted so as not to obscure the description of the various embodiments with unnecessary detail. For the purposes of the present document, the phrases “A or B” and “A/B” mean (A), (B), or (A and B).

(8) Various embodiments herein provide techniques for the UE to perform positioning measurements (e.g., on a positioning reference signal (PRS)) in a wireless cellular network. For example, embodiments may relate to the UE reporting the timing error group (TEG) information for NR positioning.

(9) With respect to the reporting requirements for the PRS measurement requirements in Rel16 [TS38.133] below, the delay requirement of DL PRS RSTD will depend on number of crucial factors which will be addressed below: the number of positioning occasions and the number of consecutive subframes, periodicity of positioning occasions, measurement sampling rate, measurement bandwidth, beam management, and other network assistance information, etc.

$$(10) T_{\text{RSTD, Total}} = \sum_{i=1}^L T_{\text{RSTD}, i} + (L - 1) * \max(T_{\text{effect}, i})$$

(11) Where, i is the index of positioning frequency layer, L is total number of positioning frequency layers, and $T_{\text{sub.effect}, i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i . $T_{\text{sub.RSTD}, i}$ is the measurement period for PRS RSTD measurement in positioning frequency layer i as specified below:

$$(12) T_{\text{RSTD}, i} = (\text{CSSF}_{\text{PRS}, i} * N_{\text{RxBeam}, i} * \text{Math.} \frac{N_{\text{PRS}, i}^{\text{slot}}}{N} \cdot \text{Math.} \frac{L_{\text{available_PRS}, i}}{N} \cdot \text{Math.} * N_{\text{sample}} - 1) * T_{\text{effect}, i} + T_{\text{last}}$$

(13) where: $N_{\text{sub.RxBeam}, i}$ is the UE Rx beam sweeping factor. In FR1, $N_{\text{sub.RxBeam}, i}=1$; and in FR2, $N_{\text{sub.RxBeam}, i}=8$. $\text{CSSF}_{\text{sub.PRS}, i}$ is the carrier-specific scaling factor for NR PRS-based positioning measurements in positioning frequency layer i as defined in clause 9.1.5.2. $N_{\text{sub.PRS.sup.slot}}$ is the maximum number of DL PRS resources in positioning frequency layer i configured in a slot.

$L_{\text{sub.available_PRS}, i}$ is the time duration of available PRS to be measured in the positioning frequency layer i , and is calculated in the same way as PRS duration K defined in clause 5.1.6.5 of TS 38.214 [26].

$N_{\text{sub.sample}}$ is the number of PRS RSTD samples and $N_{\text{sub.sample}}=4$. $T_{\text{sub.last}}$ is the measurement duration for the last PRS RSTD sample, including the sampling time and processing time,

$T_{\text{sub.last}}=T_{\text{sub}, i}+T_{\text{sub.available_PRS}, i}$, $T_{\text{sub.effect}, i}$ is the periodicity of the PRS RSTD measurement in positioning frequency layer i defined as:

$$(14) T_{\text{effect}, i} = \text{Math.} \frac{T_i}{T_{\text{available_PRS}, i}} \cdot \text{Math.} * T_{\text{available_PRS}, i}$$

(15) Where, $T_{\text{sub}, i}$ corresponds to $\text{durationOfPRS-ProcessingSymbolsInEveryTms}$ in TS 37.355 [34], $T_{\text{sub.available_PRS}, i}=\text{LCM}(T_{\text{sub.PRS}, i}, \text{MGRP}_{\text{sub}, i})$, the least common multiple between $T_{\text{sub.PRS}, i}$ and $\text{MGRP}_{\text{sub}, i}$. $\text{MGRP}_{\text{sub}, i}$ is the repetition periodicity of the measurement gap applicable for measurement in the PRS frequency layer i . $T_{\text{sub.PRS}, i}$ is the periodicity of DL PRS resource with muting on positioning frequency layer i .

(16) If more than one PRS periodicities are configured in positioning frequency layer i , the least common multiple of PRS periodicities T_{PRS} with muting among all DL PRS resource sets in the positioning frequency layer is used to derive the measurement period of that positioning frequency layer i . Where, $T_{\text{sub.per.sup.PRS with muting}}=N_{\text{sub.muting}}*T_{\text{sub.per.sup.PRS}}$ is the PRS periodicity with muting per PRS resource, $T_{\text{sub.per.sup.PRS}}$ is the periodicity of PRS resource sets given by the higher-layer parameter DL-PRS-Periodicity. $N_{\text{sub.muting}}$ is the scaling factor considering PRS resource muting. If bitmap {b.sup.1} for higher-layer parameter DL-PRS-MutingPattern is provided, and $T_{\text{sub.per.sup.PRS}}*T_{\text{sub.muting.sup.PRS}} \leq 10240$ ms, then

$$(17) N_{\text{muting}} = T_{\text{muting}}^{\text{PRS}} * \min(L, \frac{10240}{T_{\text{per}}^{\text{PRS}} * T_{\text{muting}}^{\text{PRS}}});$$

otherwise, if bitmap {b.sup.1} is not provided or $T_{\text{sub.per.sup.PRS}}*T_{\text{sub.muting.sup.PRS}} > 10240$ ms, then $N_{\text{sub.muting}}=1$. $T_{\text{sub.muting.sup.PRS}}$ is the muting repetition factor given by the higher-layer parameter

DL-PRS-MutingBitRepetitionFactor, and L is the size of the bitmap {b.sup.1}.

(18) Note: For the purpose of calculating TPRS,i, only the PRS resources fully or partially covered by the MG are considered. {N, T} is UE capability combination per band where N is a duration of DL PRS symbols in ms corresponding to durationOfPRS-ProcessingSysmbols in TS 37.355 [34] processed every T ms corresponding to durationOfPRS-ProcessingSymbolsInEveryTms in TS 37.355 [34] for a given maximum bandwidth supported by UE corresponding to supportedBandwidthPRS in TS 37.355 [34]. N' is UE capability for number of DL PRS resources that it can process in a slot as indicated by maxNumOfDL-PRS-ResProcessedPerSlot specified in TS 37.355 [34].

(19) The time T.sub.RSTD,i starts from the first MG instance aligned with a DL PRS resource(s) of positioning frequency layer i closest in time after both the NR-TDOA-ProvideAssistanceData message and NR-TDOA-RequestLocationInformation message are delivered from LMF to the physical layer of UE via LPP [34].

(20) Periodicity of PRS Occasion

(21) For the gap-less measurement, the periodicity of PRS occasion (T.sub.available_PRS,i) can be same as the configured PRS periodicity (T.sub.PRS,i) by LMF because no MGRP.

(22) The network configured T.sub.PRS is broadcasted to all UEs in a cell. Thus, according to RAN1 agreements on how to define UE processing capability ("T"), it is possible that there is misalignment between the network configured T.sub.PRS and UE specific T. As a result, from RAN4 perspective, the measurement delay requirements may depend on the larger interval of available PRSs. Thus max(T, T.sub.PRS) may be taken as the basic timing interval for PRS measurement latency requirements.

(23) In some embodiments, the basic timing interval to be used to define PRS measurement period may be:

(24)
$$T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i}$$

Applicable Number of Positioning Frequency Layers (PFLs)

(25) In some embodiments, in case of the gap-less PRS measurement, there is only single PFL to be measurement. Otherwise, the gap may be needed. But it is also feasible for the UE which can support the gap less measurement for different PFLs (e.g. the center frequency of different PFLs are same but the active BWP of them not).

(26) Accordingly, in some embodiments, the PRS gap-less measurement requirements is applicable for the single PFL only.

(27) In other embodiments, when the multiple PFLs configured, UE can decide perform the intra-f measurement for the PFL which can contained within UE's active bandwidth part (BWP). The UE may perform measurements on other PFLs by inter-frequency measurement.

(28) PRS Processing Window

(29) Ti can impact the Teffect time above.

(30) Carrier Specific Scaling Factor (CSSF)

(31) If the gap-less PRS measurement used, the other concurrent PRS processing and the legacy RRM measurements can be possible. The existing CSSF may be updated accordingly.

(32) Measurement Gap (MG) Activation and Deactivation Mechanisms

(33) In RAN1, the possible measurement gap enhancement in RAN1 were discussed. Up to now, some agreements/working assumptions in RAN1 was achieved [R1-2108639]:

(34) Subject to UE capability, support PRS measurement outside the MG, within a PRS processing window, and UE.

(35) As the pre-configured MG for PRS was introduced in Rel17, the corresponding requirements for this new aspect may be introduced. For example, in some embodiments, requirements may be defined for the case with MG switch, e.g. by taking into account the new MG.

(36) Systems and Implementations

(37) FIGS. 1-3 illustrate various systems, devices, and components that may implement aspects of disclosed embodiments.

(38) FIG. 1 illustrates a network 100 in accordance with various embodiments. The network 100 may operate in a manner consistent with 3GPP technical specifications for LTE or 5G/NR systems. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems, or the like.

(39) The network 100 may include a UE 102, which may include any mobile or non-mobile computing device designed to communicate with a RAN 104 via an over-the-air connection. The UE 102 may be communicatively coupled with the RAN 104 by a Uu interface. The UE 102 may be, but is not limited to, a

smartphone, tablet computer, wearable computer device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc.

(40) In some embodiments, the network **100** may include a plurality of UEs coupled directly with one another via a sidelink interface. The UEs may be M2M/D2D devices that communicate using physical sidelink channels such as, but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc.

(41) In some embodiments, the UE **102** may additionally communicate with an AP **106** via an over-the-air connection. The AP **106** may manage a WLAN connection, which may serve to offload some/all network traffic from the RAN **104**. The connection between the UE **102** and the AP **106** may be consistent with any IEEE 802.11 protocol, wherein the AP **106** could be a wireless fidelity (Wi-Fi®) router. In some embodiments, the UE **102**, RAN **104**, and AP **106** may utilize cellular-WLAN aggregation (for example, LWA/LWIP). Cellular-WLAN aggregation may involve the UE **102** being configured by the RAN **104** to utilize both cellular radio resources and WLAN resources.

(42) The RAN **104** may include one or more access nodes, for example, AN **108**. AN **108** may terminate air-interface protocols for the UE **102** by providing access stratum protocols including RRC, PDCP, RLC, MAC, and L1 protocols. In this manner, the AN **108** may enable data/voice connectivity between CN **120** and the UE **102**. In some embodiments, the AN **108** may be implemented in a discrete device or as one or more software entities running on server computers as part of, for example, a virtual network, which may be referred to as a CRAN or virtual baseband unit pool. The AN **108** be referred to as a BS, gNB, RAN node, eNB, ng-eNB, NodeB, RSU, TRxP, TRP, etc. The AN **108** may be a macrocell base station or a low power base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

(43) In embodiments in which the RAN **104** includes a plurality of ANs, they may be coupled with one another via an X2 interface (if the RAN **104** is an LTE RAN) or an Xn interface (if the RAN **104** is a 5G RAN). The X2/Xn interfaces, which may be separated into control/user plane interfaces in some embodiments, may allow the ANs to communicate information related to handovers, data/context transfers, mobility, load management, interference coordination, etc.

(44) The ANs of the RAN **104** may each manage one or more cells, cell groups, component carriers, etc. to provide the UE **102** with an air interface for network access. The UE **102** may be simultaneously connected with a plurality of cells provided by the same or different ANs of the RAN **104**. For example, the UE **102** and RAN **104** may use carrier aggregation to allow the UE **102** to connect with a plurality of component carriers, each corresponding to a Pcell or Scell. In dual connectivity scenarios, a first AN may be a master node that provides an MCG and a second AN may be secondary node that provides an SCG. The first/second ANs may be any combination of eNB, gNB, ng-eNB, etc.

(45) The RAN **104** may provide the air interface over a licensed spectrum or an unlicensed spectrum. To operate in the unlicensed spectrum, the nodes may use LAA, eLAA, and/or feLAA mechanisms based on CA technology with PCells/Scells. Prior to accessing the unlicensed spectrum, the nodes may perform medium/carrier-sensing operations based on, for example, a listen-before-talk (LBT) protocol.

(46) In V2X scenarios the UE **102** or AN **108** may be or act as a RSU, which may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable AN or a stationary (or relatively stationary) UE. An RSU implemented in or by: a UE may be referred to as a “UE-type RSU”; an eNB may be referred to as an “eNB-type RSU”; a gNB may be referred to as a “gNB-type RSU”; and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs. The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may provide other cellular/WLAN communications services. The components of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network.

(47) In some embodiments, the RAN **104** may be an LTE RAN **110** with eNBs, for example, eNB **112**. The

LTE RAN **110** may provide an LTE air interface with the following characteristics: SCS of 15 kHz; CP-OFDM waveform for DL and SC-FDMA waveform for UL; turbo codes for data and TBCC for control; etc. The LTE air interface may rely on CSI-RS for CSI acquisition and beam management; PDSCH/PDCCH DMRS for PDSCH/PDCCH demodulation; and CRS for cell search and initial acquisition, channel quality measurements, and channel estimation for coherent demodulation/detection at the UE. The LTE air interface may be operating on sub-6 GHz bands.

(48) In some embodiments, the RAN **104** may be an NG-RAN **114** with gNBs, for example, gNB **116**, or ng-eNBs, for example, ng-eNB **118**. The gNB **116** may connect with 5G-enabled UEs using a 5G NR interface. The gNB **116** may connect with a 5G core through an NG interface, which may include an N2 interface or an N3 interface. The ng-eNB **118** may also connect with the 5G core through an NG interface, but may connect with a UE via an LTE air interface. The gNB **116** and the ng-eNB **118** may connect with each other over an Xn interface.

(49) In some embodiments, the NG interface may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the nodes of the NG-RAN **114** and a UPF **148** (e.g., N3 interface), and an NG control plane (NG-C) interface, which is a signaling interface between the nodes of the NG-RAN **114** and an AMF **144** (e.g., N2 interface).

(50) The NG-RAN **114** may provide a 5G-NR air interface with the following characteristics: variable SCS; CP-OFDM for DL, CP-OFDM and DFT-s-OFDM for UL; polar, repetition, simplex, and Reed-Muller codes for control and LDPC for data. The 5G-NR air interface may rely on CSI-RS, PDSCH/PDCCH DMRS similar to the LTE air interface. The 5G-NR air interface may not use a CRS, but may use PBCH DMRS for PBCH demodulation; PTRS for phase tracking for PDSCH; and tracking reference signal for time tracking. The 5G-NR air interface may be operating on FR1 bands that include sub-6 GHz bands or FR2 bands that include bands from 24.25 GHz to 52.6 GHz. The 5G-NR air interface may include an SSB that is an area of a downlink resource grid that includes PSS/SSS/PBCH.

(51) In some embodiments, the 5G-NR air interface may utilize BWPs for various purposes. For example, BWP can be used for dynamic adaptation of the SCS. For example, the UE **102** can be configured with multiple BWPs where each BWP configuration has a different SCS. When a BWP change is indicated to the UE **102**, the SCS of the transmission is changed as well. Another use case example of BWP is related to power saving. In particular, multiple BWPs can be configured for the UE **102** with different amount of frequency resources (for example, PRBs) to support data transmission under different traffic loading scenarios. A BWP containing a smaller number of PRBs can be used for data transmission with small traffic load while allowing power saving at the UE **102** and in some cases at the gNB **116**. A BWP containing a larger number of PRBs can be used for scenarios with higher traffic load.

(52) The RAN **104** is communicatively coupled to CN **120** that includes network elements to provide various functions to support data and telecommunications services to customers/subscribers (for example, users of UE **102**). The components of the CN **120** may be implemented in one physical node or separate physical nodes. In some embodiments, NFV may be utilized to virtualize any or all of the functions provided by the network elements of the CN **120** onto physical compute/storage resources in servers, switches, etc. A logical instantiation of the CN **120** may be referred to as a network slice, and a logical instantiation of a portion of the CN **120** may be referred to as a network sub-slice.

(53) In some embodiments, the CN **120** may be an LTE CN **122**, which may also be referred to as an EPC. The LTE CN **122** may include MME **124**, SGW **126**, SGSN **128**, HSS **130**, PGW **132**, and PCRF **134** coupled with one another over interfaces (or “reference points”) as shown. Functions of the elements of the LTE CN **122** may be briefly introduced as follows.

(54) The MME **124** may implement mobility management functions to track a current location of the UE **102** to facilitate paging, bearer activation/deactivation, handovers, gateway selection, authentication, etc.

(55) The SGW **126** may terminate an S1 interface toward the RAN and route data packets between the RAN and the LTE CN **122**. The SGW **126** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement.

(56) The SGSN **128** may track a location of the UE **102** and perform security functions and access control. In addition, the SGSN **128** may perform inter-EPC node signaling for mobility between different RAT networks; PDN and S-GW selection as specified by MME **124**; MME selection for handovers; etc. The S3 reference point between the MME **124** and the SGSN **128** may enable user and bearer information exchange for inter-3GPP access network mobility in idle/active states.

(57) The HSS **130** may include a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The HSS **130** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **130** and the MME **124** may enable transfer of subscription and authentication data for authenticating/authorizing user access to the LTE CN **120**.

(58) The PGW **132** may terminate an SGi interface toward a data network (DN) **136** that may include an application/content server **138**. The PGW **132** may route data packets between the LTE CN **122** and the data network **136**. The PGW **132** may be coupled with the SGW **126** by an S5 reference point to facilitate user plane tunneling and tunnel management. The PGW **132** may further include a node for policy enforcement and charging data collection (for example, PCEF). Additionally, the SGi reference point between the PGW **132** and the data network **136** may be an operator external public, a private PDN, or an intra-operator packet data network, for example, for provision of IMS services. The PGW **132** may be coupled with a PCRF **134** via a Gx reference point.

(59) The PCRF **134** is the policy and charging control element of the LTE CN **122**. The PCRF **134** may be communicatively coupled to the app/content server **138** to determine appropriate QoS and charging parameters for service flows. The PCRF **132** may provision associated rules into a PCEF (via Gx reference point) with appropriate TFT and QCI.

(60) In some embodiments, the CN **120** may be a 5GC **140**. The 5GC **140** may include an AUSF **142**, AMF **144**, SMF **146**, UPF **148**, NSSF **150**, NEF **152**, NRF **154**, PCF **156**, UDM **158**, and AF **160** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the 5GC **140** may be briefly introduced as follows.

(61) The AUSF **142** may store data for authentication of UE **102** and handle authentication-related functionality. The AUSF **142** may facilitate a common authentication framework for various access types. In addition to communicating with other elements of the 5GC **140** over reference points as shown, the AUSF **142** may exhibit an Nausf service-based interface.

(62) The AMF **144** may allow other functions of the 5GC **140** to communicate with the UE **102** and the RAN **104** and to subscribe to notifications about mobility events with respect to the UE **102**. The AMF **144** may be responsible for registration management (for example, for registering UE **102**), connection management, reachability management, mobility management, lawful interception of AMF-related events, and access authentication and authorization. The AMF **144** may provide transport for SM messages between the UE **102** and the SMF **146**, and act as a transparent proxy for routing SM messages. AMF **144** may also provide transport for SMS messages between UE **102** and an SMSF. AMF **144** may interact with the AUSF **142** and the UE **102** to perform various security anchor and context management functions. Furthermore, AMF **144** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the RAN **104** and the AMF **144**; and the AMF **144** may be a termination point of NAS (N1) signaling, and perform NAS ciphering and integrity protection. AMF **144** may also support NAS signaling with the UE **102** over an N3 IWF interface.

(63) The SMF **146** may be responsible for SM (for example, session establishment, tunnel management between UPF **148** and AN **108**); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF **148** to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement, charging, and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF **144** over N2 to AN **108**; and determining SSC mode of a session. SM may refer to management of a PDU session, and a PDU session or "session" may refer to a PDU connectivity service that provides or enables the exchange of PDUs between the UE **102** and the data network **136**.

(64) The UPF **148** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to data network **136**, and a branching point to support multi-homed PDU session. The UPF **148** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform uplink traffic verification (e.g., SDF-to-QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **148** may include an uplink classifier to support routing traffic flows to a data network.

(65) The NSSF **150** may select a set of network slice instances serving the UE **102**. The NSSF **150** may also

etermine allowed NSSAI and the mapping to the subscribed S-NSSAIs, if needed. The NSSF **150** may also determine the AMF set to be used to serve the UE **102**, or a list of candidate AMFs based on a suitable configuration and possibly by querying the NRF **154**. The selection of a set of network slice instances for the UE **102** may be triggered by the AMF **144** with which the UE **102** is registered by interacting with the NSSF **150**, which may lead to a change of AMF. The NSSF **150** may interact with the AMF **144** via an N22 reference point; and may communicate with another NSSF in a visited network via an N31 reference point (not shown). Additionally, the NSSF **150** may exhibit an Nnssf service-based interface.

(66) The NEF **152** may securely expose services and capabilities provided by 3GPP network functions for third party, internal exposure/re-exposure, AFs (e.g., AF **160**), edge computing or fog computing systems, etc. In such embodiments, the NEF **152** may authenticate, authorize, or throttle the AFs. NEF **152** may also translate information exchanged with the AF **160** and information exchanged with internal network functions. For example, the NEF **152** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **152** may also receive information from other NFs based on exposed capabilities of other NFs. This information may be stored at the NEF **152** as structured data, or at a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **152** to other NFs and AFs, or used for other purposes such as analytics. Additionally, the NEF **152** may exhibit an Nnef service-based interface.

(67) The NRF **154** may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF **154** also maintains information of available NF instances and their supported services. As used herein, the terms “instantiate,” “instantiation,” and the like may refer to the creation of an instance, and an “instance” may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. Additionally, the NRF **154** may exhibit the Nnrf service-based interface.

(68) The PCF **156** may provide policy rules to control plane functions to enforce them, and may also support unified policy framework to govern network behavior. The PCF **156** may also implement a front end to access subscription information relevant for policy decisions in a UDR of the UDM **158**. In addition to communicating with functions over reference points as shown, the PCF **156** exhibit an Npcf service-based interface.

(69) The UDM **158** may handle subscription-related information to support the network entities' handling of communication sessions, and may store subscription data of UE **102**. For example, subscription data may be communicated via an N8 reference point between the UDM **158** and the AMF **144**. The UDM **158** may include two parts, an application front end and a UDR. The UDR may store subscription data and policy data for the UDM **158** and the PCF **156**, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs **102**) for the NEF **152**. The Nudr service-based interface may be exhibited by the UDR **221** to allow the UDM **158**, PCF **156**, and NEF **152** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. In addition to communicating with other NFs over reference points as shown, the UDM **158** may exhibit the Nudm service-based interface.

(70) The AF **160** may provide application influence on traffic routing, provide access to NEF, and interact with the policy framework for policy control.

(71) In some embodiments, the 5GC **140** may enable edge computing by selecting operator/3rd party services to be geographically close to a point that the UE **102** is attached to the network. This may reduce latency and load on the network. To provide edge-computing implementations, the 5GC **140** may select a UPF **148** close to the UE **102** and execute traffic steering from the UPF **148** to data network **136** via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF **160**. In this way, the AF **160** may influence UPF (re)selection and traffic routing. Based on operator deployment, when AF **160** is considered to be a trusted entity, the network operator may permit AF **160** to interact directly with relevant NFs. Additionally, the AF **160** may exhibit an Naf service-based interface.

(72) The data network **136** may represent various network operator services, Internet access, or third party services that may be provided by one or more servers including, for example, application/content server **138**.

(73) FIG. 2 schematically illustrates a wireless network **200** in accordance with various embodiments. The

wireless network **200** may include a UE **202** in wireless communication with an AN **204**. The UE **202** and AN **204** may be similar to, and substantially interchangeable with, like-named components described elsewhere herein.

(74) The UE **202** may be communicatively coupled with the AN **204** via connection **206**. The connection **206** is illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols such as an LTE protocol or a 5G NR protocol operating at mmWave or sub-6 GHz frequencies.

(75) The UE **202** may include a host platform **208** coupled with a modem platform **210**. The host platform **208** may include application processing circuitry **212**, which may be coupled with protocol processing circuitry **214** of the modem platform **210**. The application processing circuitry **212** may run various applications for the UE **202** that source/sink application data. The application processing circuitry **212** may further implement one or more layer operations to transmit/receive application data to/from a data network.

These layer operations may include transport (for example UDP) and Internet (for example, IP) operations (76) The protocol processing circuitry **214** may implement one or more of layer operations to facilitate transmission or reception of data over the connection **206**. The layer operations implemented by the protocol processing circuitry **214** may include, for example, MAC, RLC, PDCP, RRC and NAS operations.

(77) The modem platform **210** may further include digital baseband circuitry **216** that may implement one or more layer operations that are “below” layer operations performed by the protocol processing circuitry **214** in a network protocol stack. These operations may include, for example, PHY operations including one or more of HARQ-ACK functions, scrambling/descrambling, encoding/decoding, layer mapping/de-mapping, modulation symbol mapping, received symbol/bit metric determination, multi-antenna port precoding/decoding, which may include one or more of space-time, space-frequency or spatial coding, reference signal generation/detection, preamble sequence generation and/or decoding, synchronization sequence generation/detection, control channel signal blind decoding, and other related functions.

(78) The modem platform **210** may further include transmit circuitry **218**, receive circuitry **220**, RF circuitry **222**, and RF front end (RFFE) **224**, which may include or connect to one or more antenna panels **226**. Briefly, the transmit circuitry **218** may include a digital-to-analog converter, mixer, intermediate frequency (IF) components, etc.; the receive circuitry **220** may include an analog-to-digital converter, mixer, IF components, etc.; the RF circuitry **222** may include a low-noise amplifier, a power amplifier, power tracking components, etc.; RFFE **224** may include filters (for example, surface/bulk acoustic wave filters), switches, antenna tuners, beamforming components (for example, phase-array antenna components), etc. The selection and arrangement of the components of the transmit circuitry **218**, receive circuitry **220**, RF circuitry **222**, RFFE **224**, and antenna panels **226** (referred generically as “transmit/receive components”) may be specific to details of a specific implementation such as, for example, whether communication is TDM or FDM, in mmWave or sub-6 GHz frequencies, etc. In some embodiments, the transmit/receive components may be arranged in multiple parallel transmit/receive chains, may be disposed in the same or different chips/modules, etc.

(79) In some embodiments, the protocol processing circuitry **214** may include one or more instances of control circuitry (not shown) to provide control functions for the transmit/receive components.

(80) A UE reception may be established by and via the antenna panels **226**, RFFE **224**, RF circuitry **222**, receive circuitry **220**, digital baseband circuitry **216**, and protocol processing circuitry **214**. In some embodiments, the antenna panels **226** may receive a transmission from the AN **204** by receive-beamforming signals received by a plurality of antennas/antenna elements of the one or more antenna panels **226**.

(81) A UE transmission may be established by and via the protocol processing circuitry **214**, digital baseband circuitry **216**, transmit circuitry **218**, RF circuitry **222**, RFFE **224**, and antenna panels **226**. In some embodiments, the transmit components of the UE **204** may apply a spatial filter to the data to be transmitted to form a transmit beam emitted by the antenna elements of the antenna panels **226**.

(82) Similar to the UE **202**, the AN **204** may include a host platform **228** coupled with a modem platform **230**. The host platform **228** may include application processing circuitry **232** coupled with protocol processing circuitry **234** of the modem platform **230**. The modem platform may further include digital baseband circuitry **236**, transmit circuitry **238**, receive circuitry **240**, RF circuitry **242**, RFFE circuitry **244**, and antenna panels **246**. The components of the AN **204** may be similar to and substantially interchangeable with like-named components of the UE **202**. In addition to performing data transmission/reception as described above, the components of the AN **208** may perform various logical functions that include, for example, RNC functions such as radio bearer management, uplink and downlink dynamic radio resource

management and data packet scheduling.

(83) FIG. 3 is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. 3 shows a diagrammatic representation of hardware resources 300 including one or more processors (or processor cores) 310, one or more memory/storage devices 320, and one or more communication resources 330, each of which may be communicatively coupled via a bus 340 or other interface circuitry. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor 302 may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources 300.

(84) The processors 310 may include, for example, a processor 312 and a processor 314. The processors 310 may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

(85) The memory/storage devices 320 may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices 320 may include, but are not limited to, any type of volatile, non-volatile, or semi-volatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

(86) The communication resources 330 may include interconnection or network interface controllers, components, or other suitable devices to communicate with one or more peripheral devices 304 or one or more databases 306 or other network elements via a network 308. For example, the communication resources 330 may include wired communication components (e.g., for coupling via USB, Ethernet, etc.), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

(87) Instructions 350 may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors 310 to perform any one or more of the methodologies discussed herein. The instructions 350 may reside, completely or partially, within at least one of the processors 310 (e.g., within the processor's cache memory), the memory/storage devices 320, or any suitable combination thereof. Furthermore, any portion of the instructions 350 may be transferred to the hardware resources 300 from any combination of the peripheral devices 304 or the databases 306.

Accordingly, the memory of processors 310, the memory/storage devices 320, the peripheral devices 304, and the databases 306 are examples of computer-readable and machine-readable media.

Example Procedures

(88) In some embodiments, the electronic device(s), network(s), system(s), chip(s) or component(s), or portions or implementations thereof, of FIGS. 1-3, or some other figure herein, may be configured to perform one or more processes, techniques, or methods as described herein, or portions thereof.

(89) For example, FIG. 4 illustrates an example process 400 in accordance with various embodiments. The process 400 may be performed by a UE or a portion thereof. At 402, the process 400 may include identifying a positioning reference signal (PRS) measurement period based on a timing interval of:

$$(90) T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i},$$

wherein i is a positioning frequency layer index, $T_{\text{sub.effect},i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i , $T_{\text{sub},i}$ corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{\text{sub.PRS},i}$ is a periodicity of a downlink PRS resource with muting on positioning frequency layer i . At 404, the process may further include performing one or more PRS measurements within the PRS measurement period.

(91) FIG. 5 illustrates another example process 500 in accordance with various embodiments. The process 500 may be performed by a gNB or a portion thereof. At 502, the process 500 may include encoding, for transmission to a user equipment (UE), an indication of a periodicity of a downlink positioning reference signal (PRS) resource on a positioning frequency layer i , wherein a PRS measurement period is based on a timing interval of:

$$(92) T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i},$$

wherein $T_{\text{sub.effect},i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i , $T_{\text{sub},i}$ corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{\text{sub.PRS},i}$ is the

periodicity of the downlink PRS resource on the positioning frequency layer i. At 504, the process 500 may further include receiving, from the UE, one or more PRS measurements in accordance with the PRS measurement period.

(93) For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

Examples

(94) Example A1 may include one or more non-transitory computer-readable media (NTCRM) having instructions, stored thereon, that when executed by one or more processors of a user equipment (UE) configure the UE to: identify a positioning reference signal (PRS) measurement period based on a timing interval of:

$$(95) T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i},$$

wherein i is a positioning frequency layer index, $T_{\text{sub.effect},i}$ is a periodicity of a PRS measurement in a positioning frequency layer i, $T_{\text{sub},i}$ corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{\text{sub.PRS},i}$ is a periodicity of a downlink PRS resource with muting on positioning frequency layer i; and perform one or more PRS measurements within the PRS measurement period.

(96) Example A2 may include the one or more NTCRM of example A1, wherein the PRS measurement period is for a single positioning frequency layer.

(97) Example A3 may include the one or more NTCRM of example A1, wherein the instructions, when executed, are further to configure the UE to perform one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

(98) Example A4 may include the one or more NTCRM of example A1, wherein the instructions, when executed, are further to configure the UE to perform a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement without a gap.

(99) Example A5 may include the one or more NTCRM of example A1, wherein the instructions, when executed, are further to configure the UE to decode a broadcast message to determine a value of $T_{\text{sub.PRS},i}$.

(100) Example A6 may include the one or more NTCRM of example A1, wherein the one or more PRS measurements are performed without a measurement gap.

(101) Example A7 may include the one or more NTCRM of example A1, wherein the instructions, when executed, are further to configure the UE to: decode a message that activates a measurement gap for the one or more PRS measurements; and use the measurement gap upon expiration of an activation delay time.

(102) Example A8 may include one or more non-transitory computer-readable media (NTCRM) having instructions, stored thereon, that when executed by one or more processors of a next generation Node B (gNB) configure the gNB to: encode, for transmission to a user equipment (UE), an indication of a periodicity of a downlink positioning reference signal (PRS) resource on a positioning frequency layer i, wherein a PRS measurement period is based on a timing interval of:

$$(103) T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i},$$

wherein $T_{\text{sub.effect},i}$ is a periodicity of a PRS measurement in a positioning frequency layer i, $T_{\text{sub},i}$ corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{\text{sub.PRS},i}$ is the periodicity of the downlink PRS resource on the positioning frequency layer i; and receive, from the UE, one or more PRS measurements in accordance with the PRS measurement period.

(104) Example A9 may include the one or more NTCRM of example A8, wherein the PRS measurement period is for a single positioning frequency layer.

(105) Example A10 may include the one or more NTCRM of example A8, wherein the instructions, when executed, are further to configure the gNB to receive, from the UE, one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

(106) Example A11 may include the one or more NTCRM of example A8, wherein the instructions, when executed, are further to configure the gNB to receive, from the UE, a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement.

(107) Example A12 may include the one or more NTCRM of example A8, wherein the indication of the

periodicity of the downlink PRS message is transmitted via broadcast message to a plurality of UEs.

(108) Example A13 may include the one or more NTCRM of example A8, wherein the one or more PRS measurements are performed without a measurement gap.

(109) Example A14 may include the one or more NTCRM of example A8, wherein the instructions, when executed, are further to configure the gNB to: encode, for transmission to the UE, a message that activates a measurement gap for the one or more PRS measurements, wherein the UE is to use the measurement gap upon expiration of an activation delay time.

(110) Example A15 may include an apparatus to be implemented in a user equipment (UE), the apparatus comprising: a memory to store a value, $T_{\text{sub.PRS},i}$, of a periodicity of a downlink positioning reference signal (PRS) resource on a positioning frequency layer i ; and processor circuitry coupled to the memory. The processor circuitry is to: identify a positioning reference signal (PRS) measurement period based on a timing interval of:

$$(111) 0T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i},$$

wherein i is a positioning frequency layer index, $T_{\text{sub.effect},i}$ is an effective periodicity of a PRS measurement in a positioning frequency layer i , $T_{\text{sub},i}$ corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{\text{sub.PRS},i}$ is a periodicity of a downlink PRS resource with muting on positioning frequency layer i ; and perform one or more PRS measurements within the PRS measurement period.

(112) Example A16 may include the apparatus of example A15, wherein the PRS measurement period is for a single positioning frequency layer.

(113) Example A17 may include the apparatus of example A15, wherein the processor circuitry is further to perform one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

(114) Example A18 may include the apparatus of example A15, wherein the processor circuitry is further to perform a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement without a gap.

(115) Example A19 may include the apparatus of example A15, wherein the one or more PRS measurements are performed without a measurement gap.

(116) Example A20 may include the apparatus of example A15, wherein the processor circuitry is further to: decode a message that activates a measurement gap for the one or more PRS measurements; and use the measurement gap upon expiration of an activation delay time.

(117) Example B1 may include a method to define the requirements for UE positioning measurements without gap.

(118) Example B2 may include the method of example B1 or some other example herein, wherein the basic timing interval to be used to define PRS measurement period can be:

$$(119) T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i}$$

(120) Example B3 may include the method of example B1 or some other example herein, wherein the requirements are applicable for the single positioning frequency layer (PFL) measurements.

(121) Example B4 may include the method of example B1 or some other example herein, where when multiple PFLs configured, UE can decide perform the intra-f measurement for the PFL which are contained within UE's active BWP. And other PFLs by inter-f measurement.

(122) Example B5 may include the method to define the requirements for UE positioning measurements with pre-configured gap.

(123) Example B6 may include the method of example B5 or some other example herein, wherein the requirements on pre-configured MG based measurement can include the pre-configured MG activation delay

(124) Example B7 may include the method of example B5 or some other example herein, wherein the requirements on pre-configured MG based measurement can include the pre-configured MG activation/deactivation transition time.

(125) Example B8 may include a method of a UE, the method comprising:

(126) determining a positioning reference signal (PRS) measurement period based on a timing interval of:

$$(127) T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i};$$

(128) performing one or more PRS measurements in the PRS measurement period.

(129) Example B9 may include the method of example B8 or some other example herein, wherein the PRS measurement period is for a single positioning frequency layer (PFL).

- (130) Example B10 may include the method of example B8-B9 or some other example herein, further comprising determining to perform one or more intra-frequency measurements for a PFL which included in an active BWP of the UE, and/or performing PRS measurements on one or more other. PFLs by inter-frequency measurement.
- (131) Example Z01 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples A1-A20, B1-B10, or any other method or process described herein.
- (132) Example Z02 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples A1-A20, B1-B10, or any other method or process described herein.
- (133) Example Z03 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples A1-A20, B1-B10, or any other method or process described herein.
- (134) Example Z04 may include a method, technique, or process as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof.
- (135) Example Z05 may include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples A1-A20, B1-B10, or portions thereof.
- (136) Example Z06 may include a signal as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof.
- (137) Example Z07 may include a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof, or otherwise described in the present disclosure.
- (138) Example Z08 may include a signal encoded with data as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof, or otherwise described in the present disclosure.
- (139) Example Z09 may include a signal encoded with a datagram, packet, frame, segment, protocol data unit (PDU), or message as described in or related to any of examples A1-A20, B1-B10, or portions or parts thereof, or otherwise described in the present disclosure.
- (140) Example Z10 may include an electromagnetic signal carrying computer-readable instructions, wherein execution of the computer-readable instructions by one or more processors is to cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples A1-A20, B1-B10, or portions thereof.
- (141) Example Z11 may include a computer program comprising instructions, wherein execution of the program by a processing element is to cause the processing element to carry out the method, techniques, or process as described in or related to any of examples A1-A20, B1-B10, or portions thereof.
- (142) Example Z12 may include a signal in a wireless network as shown and described herein.
- (143) Example Z13 may include a method of communicating in a wireless network as shown and described herein.
- (144) Example Z14 may include a system for providing wireless communication as shown and described herein.
- (145) Example Z15 may include a device for providing wireless communication as shown and described herein.
- (146) Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

ABBREVIATIONS

(147) Unless used differently herein, terms, definitions, and abbreviations may be consistent with terms, definitions, and abbreviations defined in 3GPP TR 21.905 v16.0.0 (2019-06). For the purposes of the present document, the following abbreviations may apply to the examples and embodiments discussed herein.

(148) TABLE-US-00001 3GPP Third Generation Partnership Project 4G Fourth Generation 5G Fifth Generation 5GC 5G Core network AC Application Client ACR Application Context Relocation ACK

ACKnowledgement ACID Application Identification AF Application Function AM Acknowledged Mode AMBR Aggregate Maximum Bit Rate AMF Access and Mobility Management Function AN Access Network ANR Automatic Neighbour Relation AOA Angle of Arrival AP Application Protocol, Antenna Port, Access Point API Application Programming Interface APN Access Point Name ARP Allocation and Retention Priority ARQ Automatic Repeat Request AS Access Stratum ASP Application Service Provider ASN.1 Abstract Syntax Notation One AUSF Authentication Server Function AWGN Additive White Gaussian Noise BAP Backhaul Adaptation Protocol BCH Broadcast Channel BER Bit Error Ratio BFD Beam Failure Detection BLER Block Error Rate BPSK Binary Phase Shift Keying BRAS Broadband Remote Access Server BSS Business Support System BS Base Station BSR Buffer Status Report BW Bandwidth BWP Bandwidth Part C-RNTI Cell Radio Network Temporary Identity CA Carrier Aggregation, Certification Authority CAPEX CAPital EXpenditure CBRA Contention Based Random Access CC Component Carrier, Country Code, Cryptographic Checksum CCA Clear Channel Assessment CCE Control Channel Element CCCH Common Control Channel CE Coverage Enhancement CDM Content Delivery Network CDMA Code-Division Multiple Access CDR Charging Data Request CDR Charging Data Response CFRA Contention Free Random Access CG Cell Group CGF Charging Gateway Function CHF Charging Function CI Cell Identity CID Cell-ID (e.g., positioning method) CIM Common Information Model CIR Carrier to Interference Ratio CK Cipher Key CM Connection Management, Conditional Mandatory CMAS Commercial Mobile Alert Service CMD Command CMS Cloud Management System CO Conditional Optional CoMP Coordinated Multi-Point CORESET Control Resource Set COTS Commercial Off-The-Shelf CP Control Plane, Cyclic Prefix, Connection Point CPD Connection Point Descriptor CPE Customer Premise Equipment CPICH Common Pilot Channel CQI Channel Quality Indicator CPU CSI processing unit, Central Processing Unit C/R Command/Response field bit CRAN Cloud Radio Access Network, Cloud RAN CRB Common Resource Block CRC Cyclic Redundancy Check CRI Channel-State Information Resource Indicator, CSI-RS Resource Indicator C-RNTI Cell RNTI CS Circuit Switched CSCF call session control function CSAR Cloud Service Archive CSI Channel-State Information CSI-IM CSI Interference Measurement CSI-RS CSI Reference Signal CSI-RSRP CSI reference signal received power CSI-RSRQ CSI reference signal received quality CSI-SINR CSI signal-to-noise and interference ratio CSMA Carrier Sense Multiple Access CSMA/CA CSMA with collision avoidance CSS Common Search Space, Cell-specific Search Space CTF Charging Trigger Function CTS Clear-to-Send CW Codeword CWS Contention Window Size D2D Device-to-Device DC Dual Connectivity, Direct Current DCI Downlink Control Information DF Deployment Flavour DL Downlink DMTF Distributed Management Task Force DPDK Data Plane Development Kit DM-RS, DMRS Demodulation Reference Signal DN Data network DNN Data Network Name DNAI Data Network Access Identifier DRB Data Radio Bearer DRS Discovery Reference Signal DRX Discontinuous Reception DSL Domain Specific Language. Digital Subscriber Line DSLAM DSL Access Multiplexer DwPTS Downlink Pilot Time Slot E-LAN Ethernet Local Area Network E2E End-to-End EAS Edge Application Server ECCA extended clear channel assessment, extended CCA ECCE Enhanced Control Channel Element, Enhanced CCE ED Energy Detection EDGE Enhanced Datarates for GSM Evolution (GSM Evolution) EAS Edge Application Server EASID Edge Application Server Identification ECS Edge Configuration Server ECSP Edge Computing Service Provider EDN Edge Data Network EEC Edge Enabler Client EECID Edge Enabler Client Identification EES Edge Enabler Server EESID Edge Enabler Server Identification EHE Edge Hosting Environment EGMF Exposure Governance Management Function EGPRS Enhanced GPRS EIR Equipment Identity Register eLAA enhanced Licensed Assisted Access, enhanced LAA EM Element Manager eMBB Enhanced Mobile Broadband EMS Element Management System eNB evolved NodeB, E-UTRAN Node B EN-DC E-UTRA-NR Dual Connectivity EPC Evolved Packet Core EPDCCH enhanced PDCCH, enhanced Physical Downlink Control Channel EPRE Energy per resource element EPS Evolved Packet System EREG enhanced REG, enhanced resource element groups ETSI European Telecommunications Standards Institute ETWS Earthquake and Tsunami Warning System eUICC embedded UICC, embedded Universal Integrated Circuit Card E-UTRA Evolved UTRA E-UTRAN Evolved UTRAN EV2X Enhanced V2X F1AP F1 Application Protocol F1-C F1 Control plane interface F1-U F1 User plane interface FACCH Fast Associated Control Channel FACCH/F Fast Associated Control Channel/Full rate FACCH/H Fast Associated Control Channel/Half rate FACH Forward Access Channel FAUSCH Fast Uplink Signalling Channel FB Functional Block FBI Feedback Information FCC Federal Communications Commission FCCH Frequency Correction Channel FDD Frequency Division Duplex FDM Frequency Division Multiplex FDMA Frequency Division Multiple Access FE Front End FEC Forward Error Correction FFS For Further Study FFT Fast Fourier Transformation feLAA further enhanced

Licensed Assisted Access, further enhanced LAA FFA FN Frame Number FPGA Frame Programmable Gate Array
FR Frequency Range FQDN Fully Qualified Domain Name G-RNTI GERAN Radio Network Temporary
Identity GERAN GSM EDGE RAN, GSM EDGE Radio Access Network GGSN Gateway GPRS Support
Node GLONASS GLObal'naya NAVigatsionnaya Sputnikovaya Sistema (Engl.: Global Navigation Satellite
System) gNB Next Generation NodeB gNB-CU gNB- centralized unit, Next Generation NodeB centralized
unit gNB-DU gNB- distributed unit, Next Generation NodeB distributed unit GNSS Global Navigation
Satellite System GPRS General Packet Radio Service GPSI Generic Public Subscription Identifier GSM
Global System for Mobile Communications, Groupe Spécial Mobile GTP GPRS Tunneling Protocol GTP-
UGPRS Tunnelling Protocol for User Plane GTS Go To Sleep Signal (related to WUS) GUMMEI Globally
Unique MME Identifier GUTI Globally Unique Temporary UE Identity HARQ Hybrid ARQ, Hybrid
Automatic Repeat Request HANDO Handover HFN HyperFrame Number HHO Hard Handover HLR Home
Location Register HN Home Network HO Handover HPLMN Home Public Land Mobile Network HSDPA
High Speed Downlink Packet Access HSN Hopping Sequence Number HSPA High Speed Packet Access
HSS Home Subscriber Server HSUPA High Speed Uplink Packet Access HTTP Hyper Text Transfer Protocol
HTTPS Hyper Text Transfer Protocol Secure (https is http/1.1 over SSL, i.e. port 443) I-Block Information
Block ICCID Integrated Circuit Card Identification IAB Integrated Access and Backhaul ICIC Inter-Cell
Interference Coordination ID Identity, identifier IDFT Inverse Discrete Fourier Transform IE Information
element IBE In-Band Emission IEEE Institute of Electrical and Electronics Engineers IEI Information
Element Identifier IEIDL Information Element Identifier Data Length IETF Internet Engineering Task Force
IF Infrastructure IIOT Industrial Internet of Things IM Interference Measurement, Intermodulation, IP
Multimedia IMC IMS Credentials IMEI International Mobile Equipment Identity IMGI International mobile
group identity IMPI IP Multimedia Private Identity IMPU IP Multimedia Public identity IMS IP Multimedia
Subsystem IMSI International Mobile Subscriber Identity IoT Internet of Things IP Internet Protocol Isec IP
Security, Internet Protocol Security IP-CAN IP- Connectivity Access Network IP-M IP Multicast IPv4
Internet Protocol Version 4 IPv6 Internet Protocol Version 6 IR Infrared IS In Sync IRP Integration Reference
Point ISDN Integrated Services Digital Network ISIM IM Services Identity Module ISO International
Organisation for Standardisation ISP Internet Service Provider IWF Interworking- Function I-WLAN
Interworking WLAN Constraint length of the convolutional code, USIM Individual key kB Kilobyte (1000
bytes) kbps kilo-bits per second Kc Ciphering key Ki Individual subscriber authentication key KPI Key
Performance Indicator KQI Key Quality Indicator KSI Key Set Identifier ksp/s kilo-symbols per second KVM
Kernel Virtual Machine L1 Layer 1 (physical layer) L1-RSRP Layer 1 reference signal received power L2
Layer 2 (data link layer) L3 Layer 3 (network layer) LAA Licensed Assisted Access LAN Local Area
Network LADN Local Area Data Network LBT Listen Before Talk LCM LifeCycle Management LCR Low
Chip Rate LCS Location Services LCID Logical Channel ID LI Layer Indicator LLC Logical Link Control,
Low Layer Compatibility LMF Location Management Function LOS Line of Sight LPLMN Local PLMN
LPP LTE Positioning Protocol LSB Least Significant Bit LTE Long Term Evolution LWA LTE-WLAN
aggregation LWIP LTE/WLAN Radio Level Integration with IPsec Tunnel LTE Long Term Evolution M2M
Machine-to- Machine MAC Medium Access Control (protocol layering context) MAC Message
authentication code (security/encryption context) MAC-A MAC used for authentication and key agreement
(TSG T WG3 context) MAC-IMAC used for data integrity of signalling messages (TSG T WG3 context)
MANO Management and Orchestration MBMS Multimedia Broadcast and Multicast Service MBSFN
Multimedia Broadcast multicast service Single Frequency Network MCC Mobile Country Code MCG Master
Cell Group MCOT Maximum Channel Occupancy Time MCS Modulation and coding scheme MDAF
Management Data Analytics Function MDAS Management Data Analytics Service MDT Minimization of
Drive Tests ME Mobile Equipment MeNB master eNB MER Message Error Ratio MGL Measurement Gap
Length MGRP Measurement Gap Repetition Period MIB Master Information Block, Management
Information Base MIMO Multiple Input Multiple Output MLC Mobile Location Centre MM Mobility
Management MME Mobility Management Entity MN Master Node MNO Mobile Network Operator MO
Measurement Object, Mobile Originated MPBCH MTC Physical Broadcast CHannel MPDCCH MTC
Physical Downlink Control CHannel MPDSCH MTC Physical Downlink Shared CHannel MPRACH MTC
Physical Random Access CHannel MPUSCH MTC Physical Uplink Shared Channel MPLS MultiProtocol
Label Switching MS Mobile Station MSB Most Significant Bit MSC Mobile Switching Centre MSI
Minimum System Information, MCH Scheduling Information MSID Mobile Station Identifier MSIN Mobile
Station Identification Number MSISDN Mobile Subscriber ISDN Number MT Mobile Terminated, Mobile
Termination MTC Machine-Type Communications mMTC massive MTC, massive Machine- Type

Communication Multi-MIMO MIMO MIMO MIMO WUS MTC wake- up signal, MTC WUS NACK Negative Acknowledgement NAI Network Access Identifier NAS Non-Access Stratum, Non-Access Stratum layer NCT Network Connectivity Topology NC-JT Non- coherent Joint Transmission NEC Network Capability Exposure NE-DC NR-E-UTRA Dual Connectivity NEF Network Exposure Function NF Network Function NFP Network Forwarding Path NFPD Network Forwarding Path Descriptor NFV Network Functions Virtualization NFVI NFV Infrastructure NFVO NFV Orchestrator NG Next Generation, Next Gen NGEN-DC NG-RAN E- UTRA-NR Dual Connectivity NM Network Manager NMS Network Management System N-PoP Network Point of Presence NMIB, N-MIB Narrowband MIB NPBCH Narrowband Physical Broadcast CHannel NPDCCH Narrowband Physical Downlink Control CHannel NPDSCH Narrowband Physical Downlink Shared CHannel NPRACH Narrowband Physical Random Access CHannel NPUSCH Narrowband Physical Uplink Shared CHannel NPSS Narrowband Primary Synchronization Signal NSSS Narrowband Secondary Synchronization Signal NR New Radio, Neighbour Relation NRF NF Repository Function NRS Narrowband Reference Signal NS Network Service NSA Non-Standalone operation mode NSD Network Service Descriptor NSR Network Service Record NSSAI Network Slice Selection Assistance Information S-NNSAI Single- NSSAI NSSF Network Slice Selection Function NW Network NWUS Narrowband wake- up signal, Narrowband WUS NZP Non-Zero Power O&M Operation and Maintenance ODU2 Optical channel Data Unit-type 2 OFDM Orthogonal Frequency Division Multiplexing OFDMA Orthogonal Frequency Division Multiple Access OOB Out-of-band OOS Out of Sync OPEX OPERating EXpense OSI Other System Information OSS Operations Support System OTA over-the-air PAPR Peak-to-Average Power Ratio PAR Peak to Average Ratio PBCH Physical Broadcast Channel PC Power Control, Personal Computer PCC Primary Component Carrier, Primary CC P-CSCF Proxy CSCF PCell Primary Cell PCI Physical Cell ID, Physical Cell Identity PCEF Policy and Charging Enforcement Function PCF Policy Control Function PCRF Policy Control and Charging Rules Function PDCP Packet Data Convergence Protocol, Packet Data Convergence Protocol layer PDCCH Physical Downlink Control Channel PDCP Packet Data Convergence Protocol PDN Packet Data Network, Public Data Network PDSCH Physical Downlink Shared Channel PDU Protocol Data Unit PEI Permanent Equipment Identifiers PFD Packet Flow Description P-GW PDN Gateway PHICH Physical hybrid-ARQ indicator channel PHY Physical layer PLMN Public Land Mobile Network PIN Personal Identification Number PM Performance Measurement PMI Precoding Matrix Indicator PNF Physical Network Function PNFD Physical Network Function Descriptor PNFR Physical Network Function Record POC PTT over Cellular PP, PTP Point-to- Point PPP Point-to-Point Protocol PRACH Physical RACH PRB Physical resource block PRG Physical resource block group ProSe Proximity Services, Proximity-Based Service PRS Positioning Reference Signal PRR Packet Reception Radio PS Packet Services PSBCH Physical Sidelink Broadcast Channel PSDCH Physical Sidelink Downlink Channel PSCCH Physical Sidelink Control Channel PSSCH Physical Sidelink Shared Channel PSCell Primary SCell PSS Primary Synchronization Signal PSTN Public Switched Telephone Network PT-RS Phase-tracking reference signal PTT Push-to-Talk PUCCH Physical Uplink Control Channel PUSCH Physical Uplink Shared Channel QAM Quadrature Amplitude Modulation QCI QoS class of identifier QCL Quasi co-location QFI QoS Flow ID, QoS Flow Identifier QoS Quality of Service QPSK Quadrature (Quaternary) Phase Shift Keying QZSS Quasi-Zenith Satellite System RA-RNTI Random Access RNTI RAB Radio Access Bearer, Random Access Burst RACH Random Access Channel RADIUS Remote Authentication Dial In User Service RAN Radio Access Network RAND RANDom number (used for authentication) RAR Random Access Response RAT Radio Access Technology RAU Routing Area Update RB Resource block, Radio Bearer RBG Resource block group REG Resource Element Group Rel Release REQ REQuest RF Radio Frequency RI Rank Indicator RIV Resource indicator value RL Radio Link RLC Radio Link Control, Radio Link Control layer RLC AM RLC Acknowledged Mode RLC UM RLC Unacknowledged Mode RLF Radio Link Failure RLM Radio Link Monitoring RLM-RS Reference Signal for RLM RM Registration Management RMC Reference Measurement Channel RMSI Remaining MSI, Remaining Minimum System Information RN Relay Node RNC Radio Network Controller RNL Radio Network Layer RNTI Radio Network Temporary Identifier ROHC RObust Header Compression RRC Radio Resource Control, Radio Resource Control layer RRM Radio Resource Management RS Reference Signal RSRP Reference Signal Received Power RSRQ Reference Signal Received Quality RSSI Received Signal Strength Indicator RSU Road Side Unit RSTD Reference Signal Time difference RTP Real Time Protocol RTS Ready-To-Send RTT Round Trip Time Rx Reception, Receiving, Receiver S1AP S1 Application Protocol S1-MME S1 for the control plane S1-U S1 for the user plane S-CSCF serving CSCF S-GW Serving Gateway S-RNTI SRNC Radio Network Temporary Identity S-TMSI SAE Temporary Mobile Station

Identifier SA Standalone operation mode SAE System Architecture Evolution SAP Service Access Point
 SAPD Service Access Point Descriptor SAPI Service Access Point Identifier SCC Secondary Component
 Carrier, Secondary CC SCell Secondary Cell SCEF Service Capability Exposure Function SC-FDMA Single
 Carrier Frequency Division Multiple Access SCG Secondary Cell Group SCM Security Context Management
 SCS Subcarrier Spacing SCTP Stream Control Transmission Protocol SDAP Service Data Adaptation
 Protocol, Service Data Adaptation Protocol layer SDL Supplementary Downlink SDNF Structured Data
 Storage Network Function SDP Session Description Protocol SDSF Structured Data Storage Function SDT
 Small Data Transmission SDU Service Data Unit SEAF Security Anchor Function SeNB secondary eNB
 SEPP Security Edge Protection Proxy SFI Slot format indication SFTD Space-Frequency Time Diversity,
 SFN and frame timing difference SFN System Frame Number SgNB Secondary gNB SGSN Serving GPRS
 Support Node S-GW Serving Gateway SI System Information SI-RNTI System Information RNTI SIB
 System Information Block SIM Subscriber Identity Module SIP Session Initiated Protocol SiP System in
 Package SL Sidelink SLA Service Level Agreement SM Session Management SMF Session Management
 Function SMS Short Message Service SMSF SMS Function SMTC SSB-based Measurement Timing
 Configuration SN Secondary Node, Sequence Number SoC System on Chip SON Self-Organizing Network
 SpCell Special Cell SP-CSI-RNTI Semi- Persistent CSI RNTI SPS Semi-Persistent Scheduling SQN
 Sequence number SR Scheduling Request SRB Signalling Radio Bearer SRS Sounding Reference Signal SS
 Synchronization Signal SSB Synchronization Signal Block SSID Service Set Identifier SS/PBCH Block
 SSBRI SS/PBCH Block Resource Indicator, Synchronization Signal Block Resource Indicator SSC Session
 and Service Continuity SS-RSRP Synchronization Signal based Reference Signal Received Power SS-RSRQ
 Synchronization Signal based Reference Signal Received Quality SS-SINR Synchronization Signal based
 Signal to Noise and Interference Ratio SSS Secondary Synchronization Signal SSSG Search Space Set Group
 SSSIF Search Space Set Indicator SST Slice/Service Types SU-MIMO Single User MIMO SUL
 Supplementary Uplink TA Timing Advance, Tracking Area TAC Tracking Area Code TAG Timing Advance
 Group TAI Tracking Area Identity TAU Tracking Area Update TB Transport Block TBS Transport Block Size
 TBD To Be Defined TCI Transmission Configuration Indicator TCP Transmission Communication Protocol
 TDD Time Division Duplex TDM Time Division Multiplexing TDMA Time Division Multiple Access TE
 Terminal Equipment TEID Tunnel End Point Identifier TFT Traffic Flow Template TMSI Temporary Mobile
 Subscriber Identity TNL Transport Network Layer TPC Transmit Power Control TPMI Transmitted
 Precoding Matrix Indicator TR Technical Report TRP, TRxP Transmission Reception Point TRS Tracking
 Reference Signal TRx Transceiver TS Technical Specifications, Technical Standard TTI Transmission Time
 Interval Tx Transmission, Transmitting, Transmitter U-RNTI UTRAN Radio Network Temporary Identity
 UART Universal Asynchronous Receiver and Transmitter UCI Uplink Control Information UE User
 Equipment UDM Unified Data Management UDP User Datagram Protocol UDSF Unstructured Data Storage
 Network Function UICC Universal Integrated Circuit Card UL Uplink UM Unacknowledged Mode UML
 Unified Modelling Language UMTS Universal Mobile Telecommunications System UP User Plane UPF
 User Plane Function URI Uniform Resource Identifier URL Uniform Resource Locator URLLC Ultra-
 Reliable and Low Latency USB Universal Serial Bus USIM Universal Subscriber Identity Module USS UE-
 specific search space UTRA UMTS Terrestrial Radio Access UTRAN Universal Terrestrial Radio Access
 Network UwPTS Uplink Pilot Time Slot V2I Vehicle-to- Infrastructure V2P Vehicle-to- Pedestrian V2V
 Vehicle-to-Vehicle V2X Vehicle-to- everything VIM Virtualized Infrastructure Manager VL Virtual Link,
 VLAN Virtual LAN, Virtual Local Area Network VM Virtual Machine VNF Virtualized Network Function
 VNFFG VNF Forwarding Graph VNFFGD VNF Forwarding Graph Descriptor VNFM VNF Manager VoIP
 Voice-over-IP, Voice-over-Internet Protocol VPLMN Visited Public Land Mobile Network VPN Virtual
 Private Network Network VRB Virtual Resource Block WiMAX Worldwide Interoperability for Microwave
 Access WLAN Wireless Local Area Network WMAN Wireless Metropolitan Area Network WPAN Wireless
 Personal Area Network X2-C X2-Control plane X2-U X2-User plane XML extensible Markup Language
 XRES EXpected user RESponse XOR exclusive OR ZC Zadoff-Chu ZP Zero Power

Terminology

(149) For the purposes of the present document, the following terms and definitions are applicable to the examples and embodiments discussed herein.

(150) The term “circuitry” as used herein refers to, is part of, or includes hardware components such as an electronic circuit, a logic circuit, a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group), an Application Specific Integrated Circuit (ASIC), a field-programmable device (FPD) (e.g., a field-programmable gate array (FPGA), a programmable logic device (PLD), a complex PLD (CPLD), a

high-capacity PLD (HCPD), a structured ASIC, or a programmable SoC), digital signal processors (DSPs), etc., that are configured to provide the described functionality. In some embodiments, the circuitry may execute one or more software or firmware programs to provide at least some of the described functionality. The term “circuitry” may also refer to a combination of one or more hardware elements (or a combination of circuits used in an electrical or electronic system) with the program code used to carry out the functionality of that program code. In these embodiments, the combination of hardware elements and program code may be referred to as a particular type of circuitry.

(151) The term “processor circuitry” as used herein refers to, is part of, or includes circuitry capable of sequentially and automatically carrying out a sequence of arithmetic or logical operations, or recording, storing, and/or transferring digital data. Processing circuitry may include one or more processing cores to execute instructions and one or more memory structures to store program and data information. The term “processor circuitry” may refer to one or more application processors, one or more baseband processors, a physical central processing unit (CPU), a single-core processor, a dual-core processor, a triple-core processor, a quad-core processor, and/or any other device capable of executing or otherwise operating computer-executable instructions, such as program code, software modules, and/or functional processes. Processing circuitry may include more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. The terms “application circuitry” and/or “baseband circuitry” may be considered synonymous to, and may be referred to as, “processor circuitry.”

(152) The term “interface circuitry” as used herein refers to, is part of, or includes circuitry that enables the exchange of information between two or more components or devices. The term “interface circuitry” may refer to one or more hardware interfaces, for example, buses, I/O interfaces, peripheral component interfaces, network interface cards, and/or the like.

(153) The term “user equipment” or “UE” as used herein refers to a device with radio communication capabilities and may describe a remote user of network resources in a communications network. The term “user equipment” or “UE” may be considered synonymous to, and may be referred to as, client, mobile, mobile device, mobile terminal, user terminal, mobile unit, mobile station, mobile user, subscriber, user, remote station, access agent, user agent, receiver, radio equipment, reconfigurable radio equipment, reconfigurable mobile device, etc. Furthermore, the term “user equipment” or “UE” may include any type of wireless/wired device or any computing device including a wireless communications interface.

(154) The term “network element” as used herein refers to physical or virtualized equipment and/or infrastructure used to provide wired or wireless communication network services. The term “network element” may be considered synonymous to and/or referred to as a networked computer, networking hardware, network equipment, network node, router, switch, hub, bridge, radio network controller, RAN device, RAN node, gateway, server, virtualized VNF, NFVI, and/or the like.

(155) The term “computer system” as used herein refers to any type interconnected electronic devices, computer devices, or components thereof. Additionally, the term “computer system” and/or “system” may refer to various components of a computer that are communicatively coupled with one another. Furthermore, the term “computer system” and/or “system” may refer to multiple computer devices and/or multiple computing systems that are communicatively coupled with one another and configured to share computing and/or networking resources.

(156) The term “appliance,” “computer appliance,” or the like, as used herein refers to a computer device or computer system with program code (e.g., software or firmware) that is specifically designed to provide a specific computing resource. A “virtual appliance” is a virtual machine image to be implemented by a hypervisor-equipped device that virtualizes or emulates a computer appliance or otherwise is dedicated to provide a specific computing resource.

(157) The term “resource” as used herein refers to a physical or virtual device, a physical or virtual component within a computing environment, and/or a physical or virtual component within a particular device, such as computer devices, mechanical devices, memory space, processor/CPU time, processor/CPU usage, processor and accelerator loads, hardware time or usage, electrical power, input/output operations, ports or network sockets, channel/link allocation, throughput, memory usage, storage, network, database and applications, workload units, and/or the like. A “hardware resource” may refer to compute, storage, and/or network resources provided by physical hardware element(s). A “virtualized resource” may refer to compute, storage, and/or network resources provided by virtualization infrastructure to an application, device, system, etc. The term “network resource” or “communication resource” may refer to resources that are accessible by

computer devices/systems via a communications network. The term “system resources” may refer to any kind of shared entities to provide services, and may include computing and/or network resources. System resources may be considered as a set of coherent functions, network data objects or services, accessible through a server where such system resources reside on a single host or multiple hosts and are clearly identifiable.

(158) The term “channel” as used herein refers to any transmission medium, either tangible or intangible, which is used to communicate data or a data stream. The term “channel” may be synonymous with and/or equivalent to “communications channel,” “data communications channel,” “transmission channel,” “data transmission channel,” “access channel,” “data access channel,” “link,” “data link,” “carrier,” “radiofrequency carrier,” and/or any other like term denoting a pathway or medium through which data is communicated. Additionally, the term “link” as used herein refers to a connection between two devices through a RAT for the purpose of transmitting and receiving information.

(159) The terms “instantiate,” “instantiation,” and the like as used herein refers to the creation of an instance. An “instance” also refers to a concrete occurrence of an object, which may occur, for example, during execution of program code.

(160) The terms “coupled,” “communicatively coupled,” along with derivatives thereof are used herein. The term “coupled” may mean two or more elements are in direct physical or electrical contact with one another, may mean that two or more elements indirectly contact each other but still cooperate or interact with each other, and/or may mean that one or more other elements are coupled or connected between the elements that are said to be coupled with each other. The term “directly coupled” may mean that two or more elements are in direct contact with one another. The term “communicatively coupled” may mean that two or more elements may be in contact with one another by a means of communication including through a wire or other interconnect connection, through a wireless communication channel or link, and/or the like.

(161) The term “information element” refers to a structural element containing one or more fields. The term “field” refers to individual contents of an information element, or a data element that contains content.

(162) The term “SMTC” refers to an SSB-based measurement timing configuration configured by SSB-MeasurementTimingConfiguration.

(163) The term “SSB” refers to an SS/PBCH block.

(164) The term a “Primary Cell” refers to the MCG cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure.

(165) The term “Primary SCG Cell” refers to the SCG cell in which the UE performs random access when performing the Reconfiguration with Sync procedure for DC operation.

(166) The term “Secondary Cell” refers to a cell providing additional radio resources on top of a Special Cell for a UE configured with CA.

(167) The term “Secondary Cell Group” refers to the subset of serving cells comprising the PSCell and zero or more secondary cells for a UE configured with DC.

(168) The term “Serving Cell” refers to the primary cell for a UE in RRC_CONNECTED not configured with CA/DC there is only one serving cell comprising of the primary cell.

(169) The term “serving cell” or “serving cells” refers to the set of cells comprising the Special Cell(s) and all secondary cells for a UE in RRC_CONNECTED configured with CA/.

(170) The term “Special Cell” refers to the PCell of the MCG or the PSCell of the SCG for DC operation; otherwise, the term “Special Cell” refers to the Pcell.

Claims

1. One or more non-transitory computer-readable media (NTCRM) having instructions, stored thereon, that when executed by one or more processors of a user equipment (UE) configure the UE to: identify a positioning reference signal (PRS) measurement period based on a timing interval of:

$$T_{\text{effect}, i} = \text{Math. } \frac{T_i}{T_{\text{available_PRS}, i}} \cdot \text{Math. } * T_{\text{available_PRS}, i} \text{ wherein } i \text{ is a positioning frequency layer index,}$$

$T_{\text{sub.effect}, i}$ is a periodicity of an effective PRS measurement in a positioning frequency layer i , $T_{\text{sub.i}}$ corresponds to a duration of PRS processing symbols in every T milliseconds, $T_{\text{sub.available_PRS}, i}$ is a least common multiple between $T_{\text{sub.PRS}, i}$ and a repetition periodicity of a PRS processing window for measurement in positioning frequency layer i , and $T_{\text{sub.PRS}, i}$ is a periodicity of a downlink PRS resource

with muting on positioning frequency layer i; and perform one or more PRS measurements within the PRS measurement period.

2. The one or more NTCRM of claim 1, wherein the PRS measurement period is for a single positioning frequency layer.

3. The one or more NTCRM of claim 1, wherein the instructions, when executed, are further to configure the UE to perform one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

4. The one or more NTCRM of claim 1, wherein the instructions, when executed, are further to configure the UE to perform a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement without a gap.

5. The one or more NTCRM of claim 1, wherein the instructions, when executed, are further to configure the UE to decode a broadcast message to determine a value of $T_{\text{sub.PRS},i}$.

6. The one or more NTCRM of claim 1, wherein the one or more PRS measurements are performed without a measurement gap.

7. The one or more NTCRM of claim 1, wherein the instructions, when executed, are further to configure the UE to: decode a message that activates a measurement gap for the one or more PRS measurements; and use the measurement gap upon expiration of an activation delay time.

8. One or more non-transitory computer-readable media (NTCRM) having instructions, stored thereon, that when executed by one or more processors of a next generation Node B (gNB) configure the gNB to: encode, for transmission to a user equipment (UE), an indication of a periodicity of a downlink positioning reference signal (PRS) resource with muting on a positioning frequency layer i, wherein a PRS measurement period is based on a timing interval of: $T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{available_PRS},i}} \cdot \text{Math.} * T_{\text{available_PRS},i}$ wherein $T_{\text{sub.effect},i}$ is a periodicity of a PRS measurement in a positioning frequency layer i, $T_{\text{sub.i}}$ corresponds to a duration of PRS processing symbols in every T milliseconds, $T_{\text{sub.available_PRS},i}$ is a least common multiple between $T_{\text{sub.PRS},i}$ and a repetition periodicity of a PRS processing window for measurement in positioning is a least common multiple between $T_{\text{sub.PRS},i}$ and a repetition periodicity of a PRS processing window for measurement in positioning frequency layer i, and $T_{\text{sub.PRS},i}$ the periodicity of the downlink PRS resource with muting on the positioning frequency layer i; and receive, from the UE, one or more PRS measurements in accordance with the PRS measurement period.

9. The one or more NTCRM of claim 8, wherein the PRS measurement period is for a single positioning frequency layer.

10. The one or more NTCRM of claim 8, wherein the instructions, when executed, are further to configure the gNB to receive, from the UE, one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

11. The one or more NTCRM of claim 8, wherein the instructions, when executed, are further to configure the gNB to receive, from the UE, a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement.

12. The one or more NTCRM of claim 8, wherein the indication of the periodicity of the downlink PRS message is transmitted via broadcast message to a plurality of UEs.

13. The one or more NTCRM of claim 8, wherein the one or more PRS measurements are performed without a measurement gap.

14. The one or more NTCRM of claim 8, wherein the instructions, when executed, are further to configure the gNB to: encode, for transmission to the UE, a message that activates a measurement gap for the one or more PRS measurements, wherein the UE is to use the measurement gap upon expiration of an activation delay time.

15. An apparatus to be implemented in a user equipment (UE), the apparatus comprising: a memory to store a value, $T_{\text{sub.PRS},i}$, of a periodicity of a downlink positioning reference signal (PRS) resource with muting on a positioning frequency layer i; and processor circuitry coupled to the memory, the processor circuitry to: identify a positioning reference signal (PRS) measurement period based on a timing interval of:

$T_{\text{effect},i} = \text{Math.} \frac{T_i}{T_{\text{PRS},i}} \cdot \text{Math.} * T_{\text{PRS},i}$ wherein i is a positioning frequency layer index, $T_{\text{sub.effect},i}$ is a

periodicity of an effective PRS measurement in a positioning frequency layer i, $T_{\text{sub.i}}$ corresponds to a duration of PRS processing symbols in every T milliseconds, and $T_{\text{sub.available PRS},i}$ is a least common multiple between $T_{\text{sub.PRS},i}$ and a repetition periodicity of a PRS processing window for measurement in positioning frequency layer i, and perform one or more PRS measurements in the PRS measurement period.

16. The apparatus of claim 15, wherein the PRS measurement period is for a single positioning frequency layer.

17. The apparatus of claim 15, wherein the processor circuitry is further to perform one or more intra-frequency measurements for a positioning frequency layer which is included in an active bandwidth part of the UE.

18. The apparatus of claim 15, wherein the processor circuitry is further to perform a PRS measurement on one or more other positioning frequency layers via an inter-frequency measurement without a gap.

19. The apparatus of claim 15, wherein the one or more PRS measurements are performed without a measurement gap.

20. The apparatus of claim 15, wherein the processor circuitry is further to: decode a message that activates a measurement gap for the one or more PRS measurements; and use the measurement gap upon expiration of an activation delay time.
